

PROCESS COSTING

1. Process Costing

It is used in case of industries where output is obtained by passing through multiple process i.e. output of one process becomes the input for subsequent process until finished goods are obtained.

2. Normal Loss

It is a loss which is unavoidable in nature.

Units of such loss are shown on the credit side of process account at scrap value.

Draft Normal Loss Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process I A/c →	-	-	By Cash A/c (Process – I)*	-	-
To Process II A/c →	-	-	By Cash A/c (Process – II)	-	-
			By <u>Abnormal Gain A/c**</u>	-	-
	-	-		-	-

3. Abnormal Loss

It is a loss which is avoidable in nature.

Units of such loss are shown on the credit side of process account at NCPU.

Draft Abnormal Loss Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process I A/c →	-	-	By Cash A/c (Process – I)*	-	-
To Process II A/c →	-	-	By Cash A/c (Process – II)	-	-
			By Costing P&L A/c →	-	-
	-	-	(Balancing Figure)	-	-

* Units will be sold at scrap value only being a damaged unit.

4. Abnormal Gain

It is unexpected production during normal conditions.

Units of such gain are shown on the debit side of process account at NCPU.

Draft Abnormal Gain Account

Particulars	Units	Amount	Particulars	Units	Amount
To Normal Loss A/c*	-	-	By Process I A/c	✓ -	✓ -
To Costing P&L A/c	-	-			
(Balancing Figure)	-	-		-	-

- * The abnormal gain units valued at scrap will be shown here since only the actual profit is to be taken to costing P&L account.

5. Process Account

Draft Process Account

Particulars	Units	Amount	Particulars	Units	Amount
To Previous Process A/c →	-	-	By Normal Loss A/c (weight loss)	-	-
To Material (Σ+P) →	-	-	By Normal loss A/c (having scrap value)	-	-
To Labour →	-	-	By Normal loss A/c (requiring cost to be incurred)	-	-
To Factory OHs →	-	-	By Abnormal loss A/c	-	-
To Toxic Waste A/c	N8.123	-	By Next Process A/c (Tfd.)	-	-
To Abnormal Gain A/c	-	(NC PU)	By Costing P&L A/c (sold)	-	-
			By Finished Goods A/c	-	-

$$\text{Normal cost per unit (NCPU)} = \frac{\text{Total cost} - \text{Scrap value of Normal loss}}{\text{Total units} - \text{Normal loss units}}$$

Question – 1

Meta Company Ltd. is engaged in the production of product 'Trio' which passes through two different processes Process P and Process Q. Other information obtained from books of account for the year is as follows:

Particulars	Process P	Process Q
Raw material used →	10,000	-
Raw material cost per unit →	₹ 80	-
Direct wages →	₹ 52,000	₹ 78,000
Direct expenses →	₹ 8,600	₹ 11,100
Selling price per unit of output	₹ 130	₹ 190

Production overheads of ₹ 3,00,000 are recovered as percentage of direct wages.

$$Q = 9200 \times 3M = 6400 \quad \text{Sold} = 9200 \times 190 = 2300$$

Actual output of the two processes was:

P-9,200 units and Q-6,400 units. 3/4th of the output of Process P was passed on to the Process Q and the balance was sold. The entire output of process Q was sold.

Management & Selling expenses during the year were ₹ 1,70,000. These are not allocable to the processes.

The normal loss of the two processes, calculated on the input of every process was:

Process P- 6% and Process Q-10%

The Loss of Process P was sold at ₹ 5 per unit and that of Q at ₹ 8 per unit. Assume that Process P and Process Q are not the responsibility centres.

You are required to prepare:

- Process P Account
- Process Q Account
- Abnormal Loss and Abnormal Gain Account
- Costing Profit & Loss Account

No. 188 AK

To Pr. P	600	3000	By Bank A/c	600	3000
To Pr. Q	690	5520	By Ab. Gain	190	1520
			By Bank A/c	-	4000

Solution

(i) Process P Account

Particulars	Units	Amount	Particulars	Units	Amount
To Material	10,000	✓ 8,00,000	By Normal loss $(10000 \times 6\% \times 5)$	600	3,000
To Wages		✓ 52,000	By Process Q $(9,200 \times \frac{3}{4})$	✓ 6,900	7,17,600
To Direct expenses		✓ 8,600	By Costing Profit & Loss $(9,200 \times \frac{1}{4})$	✓ 2,300	2,39,200
To Production Overheads $(3,00,000 \times \frac{2}{5})$		✓ 1,20,000	By Abnormal loss	✓ 200	20,800
	10,000	9,80,600		10,000	9,80,600

Cost per unit = $\frac{9,80,600 - 3,000}{10,000 - 600} = ₹ 104$ per unit

(ii) Process Q Account

Particulars	Units	Amount	Particulars	Units	Amount
To Material $P.O. P A/c$	✓ 6,900	7,17,600	By Normal loss	✓ 690	5,520
To Wages		✓ 78,000	By Costing P&L	✓ 6,400	10,11,200
To Direct expenses		✓ 11,100			
To Production Overheads $(3,00,000 \times \frac{3}{5})$		✓ 1,80,000			
To Abnormal gain	190	30,020 $\rightarrow (190 \times 158)$			
	7,090	10,16,720		7,090	10,16,720

Cost per unit = $\frac{9,86,700 - 5,520}{6900 - 690} = ₹ 158$ per unit

(iii) Abnormal Loss Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process P	✓ 200	✓ 20,800	By Bank	200	1,000 $\rightarrow (200 \times 5)$
			By Costing P&L	6,400	19,800 (B/F)
	200	20,800		200	20,800

Abnormal Gain Account

Particulars	Units	Amount	Particulars	Units	Amount
To Normal loss →	190	$190 \times 8 = 1,520$	By Process Q	✓ 190	✓ 30,020
To Costing P&L		(B/F) 28,500			
	190	30,020		190	30,020

(iv)

Costing P&L Account

Particulars	Amount	Particulars	Amount
To Process P →	2,39,200	By Sales	
To Process Q →	10,11,200	P = $2,300 \times 130$	→ 2,99,000
To Abnormal loss →	19,800	Q = $6,400 \times 190$	→ 12,16,000
To Selling expenses →	1,70,000	By Abnormal gain	→ 28,500
To Net profit	1,03,300		
	15,43,500		15,43,500

Question – 2

SK Ltd. produces a product-X, which passes through three processes, I, II and III. In Process-III a by-product arises, which after further processing at a cost of ₹ 85 per unit, product Z is produced. The information related for the month of April 2021 is as follows:

	Process-I	Process-II	Process-III
Normal loss →	5%	10%	5%
Materials introduced (7,000 units) →	1,40,000	-	-
Other materials added →	62,000	1,36,000	84,200
Direct wages →	42,000	⊕ 54,000	⊕ 48,000
Direct expenses →	14,000	16,000	14,000

Production overhead for the month is ₹ 2,88,000, which is absorbed as a percentage of direct wages.

The scrapes are sold at ₹ 10 per unit

Product-Z can be sold at ₹ 135 per unit with a selling cost of ₹ 15 per unit → $135 - 15 - 85 = 35$

No. of units produced:

Process-I- 6,600; Process-II- 5,200, Process-III- 4,800 and Product-Z- 600

There is not stock at the beginning and end of the month.

You are required to prepare accounts for:

- Process-I, II and III
- By-product process.

Solution

(i)

Process- I Account

Particulars	Units	Amount(₹)	Particulars	Units	Amount(₹)
To Material	7,000	✓ 1,40,000	By Normal loss (7,000 × 5% × ₹ 10)	✓ 350	✓ 3,500
To Other Material	--	✓ 62,000	By Abnormal loss A/c (50 × ₹ 50.9022)	✓ 50	2,545
To Direct Wages	--	✓ 42,000	By Process II A/c (6,600 × ₹ 50.9022)	✓ 6,600	3,35,955
To Direct Exp.	--	✓ 14,000			
To Prod. OHs (200% × ₹ 42,000)	--	✓ 84,000			
	7,000	3,42,000		7,000	3,42,000

$$\text{Cost per unit} = \frac{3,42,000 - 3,500}{7,000 - 350} = \frac{3,38,500}{6,650} = ₹ 50.9022$$

Process- II Account

Particulars	Units	Amount(₹)	Particulars	Units	Amount(₹)
To Process-I A/c	✓ 6,600	✓ 3,35,955	By Normal loss (6,600 × 10% × ₹ 10)	✓ 660	✓ 6,600
To Other Material	--	✓ 1,36,000	By Abnormal loss A/c (740 × ₹ 108.3089)	✓ 740	✓ 80,149
To Direct Wages	--	✓ 54,000	By Process III A/c (5,200 × ₹ 108.3089)	✓ 5,200	✓ 5,63,206
To Direct Exp.	--	✓ 16,000			
To Prod. OHs (200% × ₹ 54,000)	--	✓ 1,08,000			
	6,600	6,49,955		6,600	6,49,955

$$\text{Cost per unit} = \frac{6,49,955 - 6,600}{6,600 - 660} = \frac{6,43,355}{5,940} = ₹ 108.3089$$

Process- III Account

Particulars	Units	Amount(₹)	Particulars	Units	Amount(₹)
To Process-II A/c	→ 5,200	→ 5,63,206	By Normal loss ($5,200 \times 5\% \times ₹ 10$)	260	2,600
To Other Material	--	✓ 84,200	By Product X A/c ($4,800 \times ₹ 180.1396$)	→ 4,800	8,64,870
To Direct Wages	--	✓ 48,000	By By-Product Z A/c [$600 \times (135-85-15)$]	600	21,000
To Direct Exp.	--	✓ 14,000			
To Prod. OHs ($200\% \times ₹ 48,000$)	--	✓ 96,000			
To Ab. Gain ($460 \times ₹ 180.1396$)	460	81,864			
	5,660	8,88,270		5,660	8,88,270

$$\text{Cost per unit} = \frac{8,05,406 - 2,600 - 21,000}{5,200 - 260 - 600} = \frac{7,81,806}{4,340} = ₹ 180.1396$$

(ii)

By-Product Process Account

Particulars	Units	Amount(₹)	Particulars	Units	Amount(₹)
To Process-III A/c	→ 600	21,000	By Product Z A/c	→ 600	81,000
To Processing cost (600×85)	--	51,000			
To Selling exp. (600×15)	--	9,000			
	600	81,000		600	81,000

Question – 3

A Manufacturing unit manufactures a product 'XYZ' which passes through three distinct Processes – X, Y and Z. The following data is given:

	Process X	Process Y	Process Z
Material consumed (in ₹)	→ 2,600	2,250	2,000
Direct wages (in ₹)	→ 4,000	3,500	3,000

- The total production overhead of ₹ 15,750 was recovered @ 150% of direct wages.
- 15,000 units at ₹ 2 each were introduced to process 'X'.
- The output of each process passes to the next process and finally, 12,000 units were transferred to Finished Stock Account from Process 'Z'.
- No stock of materials or work in progress was left at the end.

The following additional information is given:

Process	% of wastage to normal output	Value of Scrap per unit (₹)
X	6%	1.10
Y	?	2.00
Z	5%	1.00

You are required to:

- (i) Find out the percentage of wastage in process 'Y', given that the output of process 'Y' is transferred to Process 'Z' at ₹ 4 per unit.
- (ii) Prepare Process accounts for the three processes X, Y and Z.

Solution

- (i) Let normal loss units in process Y = y

$$\text{Normal cost per unit of Process Y} = \frac{\text{Total Cost} - \text{Scrap value of normal loss}}{\text{Total units} - \text{Normal loss unit}}$$

$$4 = \frac{52,610 - 2y}{14,100 - y}$$

$$56,400 - 4y = 52,610 - 2y$$

$$2y = 3,790$$

$$y = 1,895$$

$$\text{Thus, Normal loss \% of process Y} = \frac{1,895}{14,100} \times 100 = 13.44\%$$

- (ii) **Process X Account**

Particulars	Units	Amount	Particulars	Units	Amount
To Units Introduced	15,000	✓ 30,000	By Normal loss A/c	900	990
To Material consumed	-	✓ 2,600	($15,000 \times 6\% \times 1.10$)		
To Labour	-	✓ 4,000	By Process Y A/c	14,100	41,610
To Overheads	-	✓ 6,000	($14,100 \times 2.95106$)		
($4,000 \times 150\%$)	15,000	42,600		15,000	42,600

$$\text{Normal cost per unit} = \frac{42,600 - 990}{15,000 - 900} = \frac{41,610}{14,100} = ₹ 2.95106$$

Process Y Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process X A/c →	14,100	41,610	By Normal loss A/c	1,895	3,790
To Material consumed	-	✓ 2,250	(Part (i))		
To Labour	-	✓ 3,500	By Process Z A/c →	12,205	48,820
To Overheads	-	✓ 5,250			
($3,500 \times 150\%$)	14,100	52,610		14,100	52,610

$$\text{NC P.U.} = \frac{52,610 - (2)(y)}{14,100 - y}$$

$$4 = \frac{52,610 - 2y}{14,100 - y}$$

$$56,400 - 4y = 52,610 - 2y \Rightarrow 56,400 - 52,610 = 4y - 2y$$

$$y = \frac{3,790}{2} = 1,895$$

Process Z Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process Y A/c →	12,205	48,820	By Normal loss A/c	610	610
To Material consumed	-	2,000	(12,205 × 5% × 1)		
To Labour	-	3,000	By Finished Stock A/c	12,000	59,725
To Overheads	-	4,500	(12,000 × 4.97715)		
(3,000 × 150%)					
To Abnormal Gain A/c	405	2,015			
(405 × 4.97715)	12,610	60,335		12,610	60,335

$$\text{Normal cost per unit} = \frac{58,320 - 610}{12,205 - 610} = \frac{57,710}{11,595} = ₹ 4.97715$$

6. Treatment of Royalty / Any other Exp.

- Debit the amount of royalty on the basis of normal production units
- Excess or less payment of royalty will be adjusted in abnormal loss or gain account
- In Royalty account only final amount on actual units produced will be payable.

Question – 4

The input to a process was 16,000 kgs @ ₹ 1.20 per kg. Other expenses amounted to ₹ 2,784. The actual output from the process weighed 14,500 kgs. The normal yield of the process is 92%. Scrap is sold @ ₹ 0.50 per kg. The process is prepared under a license which provides for the royalty @ ₹ 0.15 per kg of the output. Prepare:

- Process account
- Normal wastage account
- Abnormal wastage/yield account
- Royalty account

$$\begin{aligned}
 &\text{P.L. Ac} \quad \text{Payable} \\
 &16,000 \times 92\% \times 0.15 \quad = 14,500 \times 0.15 \\
 &= 2,208 \quad = 2,175 \\
 &\quad \quad \quad \text{Ab. loss} \\
 &\quad \quad \quad = 33
 \end{aligned}$$

Solution

Process Account

Particulars	Units	Amount	Particulars	Units	Amount
To Input	16,000 ✓	✓ 19,200	By Normal loss A/c	1,280 ✓	✓ 640
To Other Exp.	-	✓ 2,784	(16,000 × 8% × 0.50)		
To Royalty	-	✓ 2,208	By Abnormal loss A/c	→ 220	✓ 352
(16,000 × 92% × 0.15)			(220 × 1.60)		
	16,000	24,192	By Finished goods A/c	14,500	✓ 23,200
			(14,500 × 1.60)	16,000	24,192

$$\text{Normal cost per unit} = \frac{24,192 - 640}{16,000 - 1,280} = \frac{23,552}{14,720} = ₹ 1.60$$

Normal Loss Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process A/c	1,280	→ 640	By Cash A/c	1,280	→ 640
	1,280	640		1,280	640

Abnormal Loss Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process A/c	220	→ 352	By Royalty A/c (220×0.15)	-	33
			By Cash A/c (220×0.50)	220	→ 110
			By Costing P&L A/c	-	209
	220	352		220	352

Royalty Account

Particulars	Amount	Particulars	Amount
To Abnormal Loss A/c	→ 33	By Process A/c	→ 2,208
To Balance c/d (14500×0.15)	2,175		
	2,208		2,208

7. Process Account with raw material stock

- Opening units of raw material stock along with its value will be debited to the process account.
- Closing units of raw material stock along with its value will be credited to the process account.
- Opening and closing stock of raw material should be adjusted while computing normal cost per unit of the process.

Normal cost per unit =

$$\frac{\text{Total Cost (includes op. raw mat.)} - \text{Closing raw mat. cost} - \text{Scrap value of Nr. loss units}}{\text{Total Units (includes op. raw mat.)} - \text{Closing raw mat. units} - \text{Normal loss units}}$$

8. Process Account with finished goods stock

- A separate process account is prepared for each process.
- Opening units of finished goods along with its value will be debited to the process stock account.
- Closing units of finished goods stock along with its value will be credited to the process stock account.
- All the goods produced by the process will be transferred to process stock account.

Question – 5

SK Ltd. processes product Z through two distinct processes – Process I and process II. On completion, it is transferred to finished stock. From the following information for the current year, prepare Process I and Process II and Finished Stock A/c.

Particulars	Process – I	Process - II
Raw materials used	7,500 units	-
Raw materials cost per unit	₹ 60	-
Transfer to next process/finished stock	7,050 units	6,525 units
Normal loss (on inputs)	5%	10%
Direct wages	→ ₹ 1,35,750	₹ 1,29,250
Direct expenses	→ 60% of direct wages	65% of direct wages
Manufacturing overheads	→ 20% of direct wages	15% of direct wages
Realisable value of scrap per unit	₹ 12.50	₹ 37.50

6,000 units of finished goods were sold at a profit of 15% on cost. Assume that there was no opening or closing stock of work-in-process.

Solution**Process I Account**

	Qty.	Amount		Qty.	Amount
To Raw material	7,500	✓ 4,50,000	By Normal Loss	375	4,688
To Direct wages		✓ 1,35,750	(5% × 7,500 × 12.5)		
To Direct expenses		✓ 81,450	By Abnormal Loss → 75	7,259	
(60% of direct wages)			(75 × 96.7947)		
To Manufacturing OHs		✓ 27,150	By Process II A/c	7,050	6,82,403
(20% of direct wages)			(7,050 × 96.7947)		
	7,500	6,94,350		7,500	6,94,350

Planned output – Process I = 7,500 – 375 = 7,125 units

Actual output = 7,050 units

Abnormal loss = (7,125 units – 7,050 units) 75 units.

Cost per unit = $\frac{6,94,350 - 4,688}{7,125} = ₹ 96.7947$

Process II Account

	Qty.	Amount		Qty.	Amount
To Process I → 7,050		6,82,403	By Normal loss (10%)	705	26,438
To Direct wages		✓ 1,29,250	(7,050 × 10% × 37.5)		
To Direct Expenses		✓ 84,013	By Finished Stock A/c	6,525	9,13,824
(65% of direct wages)			(6,525 × 140.096)		

$$(15\% \times 129250)$$

To Manufacturing OHs		✓ 19,387		
To Abnormal gain	✓ 180	✓ 25,209		
(180 × 140.096)	7,230	9,40,262	7,230	9,40,262

Planned output of Process II = 7,050 – 705 = 6,345 units

Cost per unit = $\frac{9,15,053 - 26,438}{6,345} = ₹ 140.096$

Abnormal gain = Actual output – Planned output = 6,525 – 6,345 = 180 units

Finished Stock Account

	Qty.	Amount		Qty.	Amount
To Process II →	6,525	9,13,824	By Cost of Sales A/c →	6,000 ✓	8,40,298
			By Balance c/d	525	73,526
	6,525	9,13,824	$\left[\frac{9,13,824}{6,525} \times 525 \right]$	6,525	9,13,824

Income Statement

	Amount		Amount
To Cost of Sales	8,40,298	By Abnormal Gain →	18,459
(6,000 × 140.096)		$[180 \times (140.0496 - 37.5)]$	
To Abnormal loss	→ 6,322	By Sales →	9,66,343
$[75 \times (96.7947 - 12.50)]$		$(8,40,298 \times 115\%)$	
To Net Profit (B/f)	1,38,182		
	9,84,802		9,84,802

9. Process Account with WIP stock

- Opening units of WIP along with its value will be debited to the process account.
- Closing units of WIP along with its value will be credited to the process account.

10. Valuation of WIP

- Calculate equivalent units of production for each element of cost i.e. material, labour and overheads by preparing statement of equivalent units.
- Calculate cost per equivalent unit for each element of cost i.e. material, labour and overheads.
- Calculate the value of WIP by multiplying the equivalent units of WIP along with cost per equivalent unit.

IIP	OIP	Mat.	Labr.	Org.
		1.1	1.1	1.1
Op. WIP to FG	✓			
Intro. & Comp.	✓			
Comp. & Top.	✓			
NB. lds	✓			
Ab. lds	✓			
Cl. WIP	✓			

11. Methods of WIP Stock Valuation

Default

Apply

- ① Does. Mention to use
② DOC of op. WIP not Given

	<u>FIFO Method</u>	<u>W. Average Method</u>
Ist Process		
Opening WIP	<u>Bal. DOC</u>	<u>100%</u>
Introduced & Complete	<u>100%</u>	<u>100%</u>
Normal Loss	<u>NIL</u>	<u>NIL</u>
Abnormal Loss	<u>DOC / 100%</u>	<u>DOC / 100%</u>
Closing WIP	<u>DOC</u>	<u>DOC</u>
	<u>FIFO Method</u>	<u>W. Average Method</u>
Subsequent Process or		
Double Material Questions	Prev. Per. T <u>M1</u> <u>M2, L, OH</u>	<u>M1</u> <u>M2, L, OH</u>
Opening WIP	<u>0%</u> <u>DOC</u>	<u>100%</u> <u>100%</u>
Introduced & Complete	<u>100%</u> <u>100%</u>	<u>100%</u> <u>100%</u>
Normal Loss	<u>NIL</u> <u>NIL</u>	<u>NIL</u> <u>NIL</u>
Abnormal loss	<u>100%</u> <u>DOC / 100%</u>	<u>100%</u> <u>DOC / 100%</u>
Closing WIP	<u>100%</u> <u>DOC</u>	<u>100%</u> <u>DOC</u>

↓
In w. Avg. Method, Cost of op. WIP will be added with current period cost to calculate cost per equiv. unit.

- ⊛ Ab. Gain units will be shown as -ve at 100% DOC.
- ⊛ Scrap value of Nr. loss will be adjusted from Mat. while computing cost per equiv. unit.
- ⊛ SV of Nr. loss will be adjusted from M-1 if there are two material.

Question – 6

With the help of the following information, prepare Process Account, giving full working notes:

Opening stock of work in progress: 1,000 units at ₹ 10,000

Degree of completion: Material 100%, Labour 50%, Overhead 40%

Introduced during the process: 10,000 units at ₹ 37,800

Wages: ₹ 17,840

Overheads: ₹ 8,840

Scrap 1,500 units

Degree of Completion: Materials 100%, Labour 80%, Overheads 60%

Closing work in progress: 1,000 units

Degree of completion: Materials 100%, Labour 60%, Overheads 50%

Normal loss 10% of total input

Scrap value ₹ 2 per unit

Solution**Statement of Equivalent Production**

Input	Output	Material		Labour		Overheads	
		%	Units	%	Units	%	Units
Op. WIP 1,000	Op. WIP 1,000	-	-	50	500	60	600
Input 10,000	Introduced & Complete (B/F) 7,500	100	7,500	100	7,500	100	7,500
	Transferred (B/F) 8,500						
	Normal Loss → 1,100 (11,000 × 10%)	-	-	-	-	-	-
	Abnormal Loss → 400 (1500 - 1100)	100	400	80	320	60	240
	Closing WIP → 1,000	100	1,000	60	600	50	500
→ 11,000	→ 11,000		8,900		8,920		8,840

Statement of Cost per Equivalent Unit

Particulars	Material	Labour	Overheads
Cost	✓ 37,800	✓ 17,840	✓ 8,840
Less: Normal Scrap (1,100 × 2)	(2,200)	-	-
Total →	35,600	17,840	8,840
Equivalent Units	8,900	8,920	8,840
Cost per equivalent unit	4	2	1

Statement of distribution of cost

Particulars	Element of Cost	Equivalent units	Cost per unit	Cost	Total Cost
Opening WIP to FG	Material	-	4	-	
	Labour	500	2	1,000	
	Overheads	600	1	600	1,600
Introduced & Complete	Material	7,500	4	30,000	
	Labour	7,500	2	15,000	
	Overheads	7,500	1	7,500	52,500
Abnormal Loss	Material	400	4	1,600	
	Labour	320	2	640	
	Overheads	240	1	240	2,480
Closing WIP	Material	1,000	4	4,000	
	Labour	600	2	1,200	
	Overheads	500	1	500	5,700

→ Current Pdt Loss

From Dues.

Process Account

Particulars	Units	Amount	Particulars	Units	Amount
To Opening WIP	✓ 1,000	✓ 10,000	By Normal Loss A/c	✓ 1,100	✓ 2,200
To Material	✓ 10,000	✓ 37,800	By Abnormal loss A/c	✓ 400	✓ 2,480
To Labour	-	✓ 17,840	By Next Process A/c	✓ 8,500	✓ 64,100
To Overheads	-	✓ 8,840	(52,500+1,600+10,000)		
			By closing WIP	✓ 1,000	✓ 5,700
	11,000	74,480		11,000	74,480

Question – 7

The following data is related to Process S for August:

- (a) Opening WIP → 8,000 units
 Material (100%) → ₹ 63,900
 Labour (60%) → ₹ 10,800 → 80,100
 Overheads (60%) → ₹ 5,400
- (b) Input of 1,82,000 units → ₹ 7,56,900
- (c) Labour paid → ₹ 3,28,000
- (d) Overheads incurred → ₹ 1,64,000
- (e) Units scrapped → 14,000 units
 Degree of Completion: Material 100%; Labour & Overheads 50%
- (f) Closing WIP → 18,000 units
 Degree of Completion: Material 100%; Labour & Overheads 70%
- (g) 1,58,000 units were completed and transferred to next process
- (h) Normal loss is 8% of total input
- (i) Scrap value is ₹ 8 per unit

You are required to prepare Process A/c and other relevant account assuming average method is used.

Solution**Statement of Equivalent Production**

Input	Output	Material		Labour		Overheads	
		%	Units	%	Units	%	Units
Op. WIP ✓ 8,000	Complete & Transferred → 1,58,000	100	1,58,000	100	1,58,000	100	1,58,000
Input ✓ 1,82,000	Normal Loss → 15,200 (1,90,000 × 8%)	-	-	-	-	-	-
	Abnormal Gain → (1,200) (14,000 - 15,200)	100	(1,200)	100	(1,200)	100	(1,200)
	Closing WIP → 18,000	100	18,000	70	12,600 12,600	70	12,600 12,600
→ 1,90,000	→ 1,90,000		1,74,800		1,69,400		1,69,400

Statement of Cost per Equivalent Unit

Particulars	Material	Labour	Overheads
Cost →	7,56,900	3,28,000	1,64,000
Add: Opening WIP cost →	63,900	10,800	5,400
Less: Normal Scrap (15,200 × 8) →	(1,21,600)	-	-
Total →	6,99,200	3,38,800	1,69,400
Equivalent Units →	1,74,800	1,69,400	1,69,400
Cost per equivalent unit	4	2	1

Cost of next process account = $1,58,000 \times (4 + 2 + 1) = ₹ 11,06,000$

Cost of closing WIP = $(18,000 \times 4) + (12,600 \times 2) + (12,600 \times 1) = ₹ 1,09,800$

Cost of abnormal gain = $1,200 \times (4 + 2 + 1) = ₹ 8,400$

Process S Account

Particulars	Units	Amount	Particulars	Units	Amount
To Opening WIP → 8,000	✓ 8,000	✓ 80,100	By Normal Loss A/c → 15,200	15,200	1,21,600
To Material → 1,82,000	✓ 1,82,000	✓ 7,56,900	By Next Process A/c → 1,58,000	1,58,000	11,06,000
To Labour → -	✓ -	✓ 3,28,000	By closing WIP → 1,800	1,800	1,09,800
To Overheads → -	✓ -	✓ 1,64,000			
To Abnormal Gain 1,200	1,200	8,400			
	1,91,200	13,37,400		1,91,200	13,37,400

Abnormal Gain Account

Particulars	Units	Amount	Particulars	Units	Amount
To Normal loss A/c → 1,200 (1,200 × 8)	1,200	9,600	By Process S A/c → 1,200	1,200	8,400
			By Costing P&L A/c	-	1,200 (B/F)
	1,200	9,600		1,200	9,600

Normal Loss Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process S A/c	15,200	1,21,600	By Abnormal Gain A/c	1,200	9,600
			By Bank A/c	14,000	1,12,000
	15,200	1,21,600		15,200	1,21,600

Question – 8

A company produces a component, which passes through two processes. During the month of April, materials for 40,000 components were put into Process I of which 30,000 were completed and transferred to Process II. Those not transferred to Process II were 100% complete as to materials cost and 50% complete as to labour and overheads cost. The Process I cost incurred were as follows:

Direct Materials	→ ₹ 15,000
Direct Wages	→ ₹ 18,000
Factory Overheads	→ ₹ 12,000

Of those transferred to Process II, 28,000 units were completed and transferred to finished goods stores. There was a normal loss with no salvage value of 200 units in Process II. There were 1,800 units, remained unfinished in the process with 100% complete as to materials and 25% complete as regard to wages and overheads.

No further process material costs occur after introduction at the first process until the end of the second process, when protective packing is applied to the completed components. The process and packing costs incurred at the end of the Process II were:

Packing Material	→ ₹ 4,000	For Fh only
Direct Wages	→ ₹ 3,500	
Factory Overheads	→ ₹ 4,500	

Required:

- Prepare statement of equivalent production, cost per unit and Process I A/c.
- Prepare statement of equivalent production, cost per unit and Process II A/c.

Solution

Statement of Equivalent Production and Cost per Unit for Process I

Input	Output	Material		Labour		Overheads	
		%	Units	%	Units	%	Units
Input → 40,000	Introduced & → 30,000	100	30,000	100	30,000	100	30,000
	Complete						
	Closing WIP → 10,000	100	10,000	50	5,000	50	5,000
40,000	40,000		→ 40,000		→ 35,000		→ 35,000
	Cost		→ 15,000		→ 35,000		→ 12,000
	Cost per unit		0.375		0.514286		0.342857

Statement of apportionment of cost for Process I

Particulars	Element of Cost	Equivalent units	Cost per unit	Cost	Total Cost
Introduced & Complete	Material	30,000	0.375	11,250	36,965
	Labour	30,000	0.514286	15,429	
	Overheads	30,000	0.342857	10,286	
Closing WIP	Material	10,000	0.375	3,750	8,035
	Labour	5,000	0.514286	2,571	
	Overheads	5,000	0.342857	1,714	

Process I Account

Particulars	Units	Amount	Particulars	Units	Amount
To Material	40,000	15,000	By Process II A/c	30,000	36,965
To Labour	-	18,000	By Closing WIP	10,000	8,035
To Overheads	-	12,000			
	40,000	45,000		40,000	45,000

Statement of Equivalent Production and cost per unit for Process II

Input	Output	(P. I) Material - 1		Material - 2		Labour		Overheads	
		%	Units	%	Units	%	Units	%	Units
Input 30,000	Introduced & Complete → 28,000	100	28,000	100	28,000	100	28,000	100	28,000
	Normal Loss → 200	-	-	-	-	-	-	-	-
	Closing WIP → 1,800	100	1,800	-	-	25	450	25	450
30,000	30,000		29,800		28,000		28,450		28,450
	Cost → 36,965				4,000		3,500		4,500
	Cost per unit		1.2404		0.1429		0.123		0.1582

Statement of apportionment of cost for Process II

Particulars	Element of Cost	Equivalent units	Cost per unit	Cost	Total Cost
Introduced & Complete	Material - 1	28,000	1.2404	34,732	46,606
	Material - 2	28,000	0.1429	4,000	
	Labour	28,000	0.123	3,445	
	Overheads	28,000	0.1582	4,429	

Closing WIP	Material – 1	1,800	1.2404	2,233	
	Material – 2	-	0.1429	-	
	Labour	450	0.123	55	
	Overheads	450	0.1582	71	2,359

Process II Account

Particulars	Units	Amount	Particulars	Units	Amount
To Process I A/c	30,000	36,965	By Normal loss A/c	200	-
To Packing Material	-	4,000	By Finished Goods A/c	28,000	46,606
To Labour	-	3,500	By Closing WIP	1,800	2,359
To Overheads	-	4,500			
	30,000	48,965		30,000	48,965

12. Inter Process Profit

Transfer goods from one process to other on cost plus profit basis to evaluate performance of each process separately as they are considered responsibility centre.

Question – 9

KT Ltd. produces a product EMM which passes through two processes before it is completed and transferred to finished stock. The following data relate to May 2019.

Particulars		<u>Process A</u>	<u>Process B</u>	<u>Finished Stock</u>
Opening stock	→	₹ 5,000	✓ ₹ 5,500	₹ 10,000
Direct materials	→	9,000	✓ 9,500	
Direct wages	→	5,000	✓ 6,000	
Factory overheads	→	4,600	✓ 2,030	
Closing stock	→	2,000	✓ 2,490	5,000
Inter process profit included in opening stock	→	0	✓ 1,000	4,000

Output of Process A is transferred to Process B at 25% profit on the transfer price and output of Process B is transferred to finished stock at 20% profit on the transfer price. Stock in process is valued at prime cost. Finished stock is valued at the price at which it is received from Process B. Sales during the period are ₹ 75,000.

Prepare the process cost accounts and Finished stock account showing the profit element at each stage.

<u>Pr. B ATC</u>						
To Pr. A	21600	7200	28800	By Fk ATC		
To Op.	4500	1000	5500	By Q-St.	2080	410
To DM	9500	-	9500			2490
To DL	6000	-	6000			
To F.OH	2030	-	2030			
To Profit	-	12335	12335			
			61675			

$$\text{Pr. in} = \frac{8200}{49800} \times 2490 = 410$$

Solution**Process A Account**

Particulars	Cost	Profit	Total	Particulars	Cost	Profit	Total
To Opening stock	5,000	-	5,000	By Process B A/c	28,800 21,600	7,200	21,600 28,800
To Direct material	9,000	-	9,000				
To Direct wages	5,000	-	5,000				
	19,000	-	19,000				
(-) Closing stock	(2,000)	-	(2,000)				
	17,000	-	17,000				
To Factory OHs	4,600	-	4,600				
	21,600	-	21,600				
To Profit	-	7,200	7,200				
	21,600	7,200	28,800		21,600	7,200	28,800

→ $\frac{(21,600 - 2,000)}{75\%}$

Process B Account

Particulars	Cost	Profit	Total	Particulars	Cost	Profit	Total
To Opening stock	4,500	1,000	5,500	By F. Stock A/c	41,550	20,125	<u>61,675</u>
To Process A A/c	21,600	7,200	28,800				
To Direct material	9,500	-	9,500				
To Direct wages	6,000	-	6,000				
	41,600	8,200	49,800				
(-) Closing stock	<u>(2,080)</u>	<u>(410)</u>	<u>(2,490)</u>				
	39,520	7,790	47,310				
To Factory OHs	2,030	-	2,030				
	41,550	7,790	<u>49,340</u>				
To Profit	-	12,335	<u>12,335</u>				
	41,550	20,125	<u>61,675</u>		41,550	20,125	61,675

Profit element in closing stock = $\frac{8,200}{49,800} \times 2,490 = ₹ 410$

Finished Stock Account

Particulars	Cost	Profit	Total	Particulars	Cost	Profit	Total
To Opening stock →	<u>6,000</u>	<u>4,000</u>	<u>10,000</u>	By Costing P&L A/c	44,182	30,818	<u>75,000</u>
To Process B A/c →	41,550	20,125	61,675				
	47,550	24,125	71,675				
(-) Closing stock	(3,368)	<u>(1,632)</u>	(5,000)				
	44,182	22,493	<u>66,675</u>				
To Profit (Bal. fig)	-	8,325	<u>8,325</u>				
	44,182	30,818	<u>75,000</u>		44,182	30,818	75,000

→ $\frac{20,125}{61,675} \times 5,000 = 1,632$

(21%)

$$\text{Profit element in closing stock} = \frac{20,125}{61,675} \times 5,000 = ₹ 1,632$$

Question – 10

Arnav Ltd. manufactures chemical solutions used in paint and adhesive products. Chemical solutions are produced in different processes. Some of the processes are hazardous in nature which may result in fire accidents.

At the end of the last month, one fire accident occurred in the factory. The fire destroyed some of the paper files containing records of the process operations for the month.

You being an associate to the Chief Manager (finance), are assigned to prepare the process accounts for the month during which the fire occurred. From the documents and files of other sources, following information could be retrieved:

Opening work-in-process at the beginning of the month was 500 litres, 80% complete for labour and 60% complete for overheads. Opening work-in-process was valued at ₹ 2,78,000.

Closing work-in-process at the end of the month was 100 litres, 20% complete for labour and 10% complete for overheads.

Normal loss is 10% of input (fresh) and total losses during the month were 800 litres partly due to the fire damage.

Output transferred to finished goods was 3,400 litres.

Losses have a scrap value of ₹ 20 per litre.

All raw material are added at the commencement of the process.

The cost per equivalent unit is ₹ 660 for the month made up as follows:

Raw material ₹ 300; labour ₹ 200; Overheads ₹ 160.

The company uses FIFO method to value work-in-process and finished goods. The following information are required for managerial decisions:

Question – 1

How much quantity of raw material introduced during the month?

- (a) 4,300 litres
- (b) 3,500 litres
- (c) 4,200 litres
- ✓ (d) 3,800 litres

$$\begin{aligned} \text{IIP} &= \text{OIP} \\ \text{Op. WIP} + \text{Introd.} &= \text{Comp.} + \text{Total loss} + \text{U. WIP} \\ \text{300} + \text{Introd.} &= 3400 + 800 + 100 \\ \text{Introd.} &= 3800 \end{aligned}$$

	OIP	M	L	OH
Op. WIP	500	-	-	-
I & C	2900	100%	2900	100%
C & T	3400	-	-	-
NR. loss	380	100%	420	100%
Ab. loss	420	100%	20	10%
CR WIP	100	-	-	-
	4300	3420	3440	3520
		x 300	x 200	x 160
		= 10260	= 6880	= 564800
		(+) SV NR. loss (380 x 20)	7600	-
		103600	6880	564800

Question – 2

The quantity of normal loss and abnormal loss are:

- $10\% \times 3800 = 380$
 $800 - 380 = 420$
- (a) Normal loss – 380 litres and abnormal loss – 420 litres
 - (b) Normal loss – 350 litres and abnormal loss – 450 litres
 - (c) Normal loss – 430 litres and abnormal loss – 370 litres
 - (d) Normal loss – 420 litres and abnormal loss – 380 litres

Question – 3

Value for raw material added to the process during the month is:

- (a) ₹ 10,10,000
- (b) ₹ 10,33,600
- (c) ₹ 10,18,400
- (d) ₹ 10,20,000

Question – 4

Value of labour and overhead in closing Work-in-process are:

- $(20 \times 200) = 4000$
 $(10 \times 160) = 1600$
- (a) ₹ 4,000 & ₹ 1,600 respectively
 - (b) ₹ 20,000 & ₹ 16,000 respectively
 - (c) ₹ 16,000 & ₹ 9,000 respectively
 - (d) ₹ 13,200 & ₹ 6,600 respectively

Question – 5

Value of output transferred to finished goods is:

- (a) ₹ 22,57,200
- (b) ₹ 20,06,400
- (c) ₹ 22,44,000
- (d) ₹ 19,27,200

$$\begin{aligned}
 & \text{Int. \& Comp.} \\
 & [2900 \times (300 + 200 + 160)] + [(100 \times 200) + (200 \times 160)] \\
 & + 278000 \\
 & \quad \quad \quad \text{Current op. WIP} \\
 & \quad \quad \quad \text{(LY op. WIP)} \\
 & = 22.44 \text{ P}
 \end{aligned}$$

1	2	3	4	5
D	A	B	A	C